

NG SHU JIE

Firmware / Embedded Systems / IC Design Engineer

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PROFESSIONAL SUMMARY

Computer Engineering graduate (CGPA 3.4150, UTAR) with hands-on experience across embedded systems, digital hardware design, and production-level hardware testing. Proficient in C/C++, Python, and SystemVerilog, with proven project work on RISC-V and MIPS processor design, IoT sensor systems, and STM32 embedded platforms. Completed industrial training as a Production Intern at FootFallCam performing hardware testing and debugging test scripts for newly assembled devices. Eager to learn and contribute to firmware development, embedded system development and IC design industry.

TECHNICAL SKILLS

Languages: C/C++, Python, SystemVerilog / Verilog, Assembly (MIPS, RISC-V)

Microcontrollers & Peripherals: STM32, Arduino, UART, I2C, GPIO, Timers, Interrupts

Hardware / Software Interfacing: Register-level Programming, Memory Mapping, ADC/DAC

Software & Tools: Linux Environment, Git, Keil μ Vision, Vivado, ModelSim

Core Concepts: Computer Architecture, Memory Management, FTL Concepts, Multi-threading

WORK EXPERIENCE

Production Intern – Hardware Testing

Oct 2025 – Jan 2026

FootFallCam (Meta Research Sdn Bhd) | Sungai Long, Kajang

- Awarded Top 10% Internship Certificate — the highest performance recognition given to interns at FootFallCam.
- Performed systematic hardware testing on newly assembled footfall-counting devices to verify readiness for downstream calibration and functional testing, identifying and logging defects at the production stage.
- Executed calibration scripts and production test routines via terminal; troubleshoot and debugged Python-based testing scripts to resolve logic errors and ensure smooth, uninterrupted test runs.
- Diagnosed hardware faults on assembled devices and carried out hands-on repairs, including microsoldering work on RGB LED boards and firmware flashing for microcontroller boards using terminal commands.
- Escalated complex device issues to the device engineering team through structured communication, enabling timely resolution of critical production blockers and maintaining production flow.
- Participated in daily Scrum stand-ups to report progress, flag blockers, and align on daily priorities — contributing to an Agile production workflow.

PROJECTS

Room Monitoring & Automation System

Jun 2024 – Sep 2024

Team Project (2 Members) | STM32F429i-DISC1, C, Keil μ Vision3, STLink

- Developed an STM32-based embedded system in C to monitor room temperature and humidity, reading ADC-connected LDR sensors and driving onboard LCD output with controlled feedback.
- Implemented a secure login interface with animated graphics and button-sequence authentication, covering full embedded workflow: sensor acquisition, data processing, and GUI output.
- Debugged hardware-software integration issues using STLink and Keil μ Vision3, gaining hands-on experience with embedded development and flashing cycle.

MIPS Multi-Cycle Processor Design

Jun 2025 – Sep 2025

Team Project (3 Members) | SystemVerilog, Vivado

- Architected modular RTL components (Datapath, Control Unit, ALU, Memory, CP0) for a multi-cycle MIPS processor in SystemVerilog, with full instruction set support including arithmetic, branching, memory access, and exception handling.
- Generated MIPS assembly test programs from C/C++ source for system-level validation; debugged timing and functional mismatches via waveform analysis in Vivado Simulator.
- Documented module interfaces and verification outcomes for reproducibility; also led RTL design of a 5-stage pipelined RISC-V CPU in a prior project.

EEG-Based Music Genre Classification – Final Year Project

Jun 2025 – May 2026

Individual Research Project | Python, PyTorch, scikit-learn, MNE-Python, Optuna

- Built an end-to-end EEG signal processing and Machine Learning / Deep Learning pipeline for music genre recognition (MUSIN-G dataset, 20 subjects, 4 genres), spanning preprocessing, feature extraction (raw EEG, Welch PSD, bandpower), and model training.
- Identified and quantified data leakage in earlier results - random k-fold CV inflated RF accuracy to 90.7%; redesigned evaluation with strict subject-wise splitting method (14/3/3 train/val/test), achieving a leakage-free 47.50% accuracy - nearly double the 25% chance baseline.
- Benchmarked 96 pipeline configurations across 7 model architecture (SVM, Random Forest, XGBoost; CNN, CNN-LSTM, CNN-BiLSTM, CNN-GRU) with Optuna TPE hyperparameter optimisation.
- Validated real-world practicality using self-recorded data from consumer EEG headset (Emotiv EPOC+).

EDUCATION

Bachelor of Information Technology (Honours) – Computer Engineering

Jun 2023 – Jun 2026

Universiti Tunku Abdul Rahman (UTAR), Kampar, Perak

CGPA: 3.4150

Relevant Coursework:

- Embedded Systems Design, Processor & Interfacing, Advanced Computer Architecture, Data Structures & Algorithmic Problem Solving, Operating Systems, Digital Systems Design, Programming Concepts & Practices

Languages Spoken:

- Mandarin Chinese (Native) | English (Fluent) | Malay (Intermediate) | Japanese (Basic)
- MUET Band 4.5

LEADERSHIP & EXTRACURRICULAR

Finance Leader – "Start Managing Your Personal Finance" Talk

Nov 2024

Universiti Tunku Abdul Rahman (UTAR), Kampar, Perak

- Spearheaded planning, budgeting, and logistics for a campus-wide financial literacy event attracting 70+ student attendees.
- Coordinated with external financial professionals to deliver talks on investment, budgeting, and career finance planning.
- Demonstrated stakeholder coordination, time management, and team leadership under tight planning timelines.